

Ari Gestetner

 github.com/ariges770 |  gestetnerari@gmail.com |  [GitHub](https://github.com)  linkedin.com/ari-gestetner

Education

Monash University | Melbourne, Victoria, Australia

Bachelor Commerce and Science in Mathematical Foundations of Econometrics, Mathematics and Computational Science

2023-2025

- Awarded with summer and winter vacation research scholarships
- First place in an IMC algorithmic trading competition for Monash students hosted by Monash Association of Coding (MAC)

Experience

Teaching Associate

Feb 2024 - Present

Monash University

- Achieved 89% student satisfaction rating exceeding the faculty average of 82% by providing engaging, individualised and interactive classes and explanations
- Collaborated with fellow TAs to deliver applied classes, sharing feedback to drive continuous teaching improvements and enhance student learning outcomes

Research Scholarships

Monash University

Nov 2024 - Feb 2025 - Dr. Buser Say

Python, SCIP, Constraint Programming

- Continued extending SCIPPlan to allow users to provide a system of odes to be solved and used in the planning process supporting engineers and physicists in their work
- Incorporated function interpolation, introducing flexibility to the planning process and allowing for more complex problems to be solved

Jul 2024 - Dr. Buser Say

Python, SCIP, Constraint Programming

- Developed SCIPPlan+, an automated planner for discrete and continuous time planning problems with stochastic domains
- Researched the applications and performance of SCIPPlan+ over a variety of domains

Nov 2023 - Feb 2024 - Dr. Buser Say

Python, SCIP, Constraint Programming

- Refactored and extended [SCIPPlan](#) an automated planner for continuous time sequential planning problems over non-linear domains
- Assessed the performance of SCIPPlan in identifying optimal plans by formulating diverse problems with non-linear constraints
- Successfully derived feasible optimal solutions for a novel problem within a reasonable time frame

Jul 2023 - Dr. Buser Say

Python, Cplex, Gurobi, Seaborn, MILP

- Investigated various linear programming constraint formulations based on neural network transition functions
- Benchmarked and presented different solvers and formulations finding results of over 2x decrease in solve time

Projects

Pandemic Prediction & Planning Python, SciPy, Matplotlib

- Explored complex disease spread dynamics by extending the standard SIR model to include real-world factors like birth and death rates to answer specific modelling questions.
- Utilised stochastic systems to create prediction intervals for possible outcomes and applied genetic algorithms to identify optimal lockdown schedules that minimise duration while keeping infections below a safe cap.

Retail Sales Forecasting R, Fable, Tidyverse

- Analysed multi-decade Queensland retail turnover data to identify long-term growth trends and predictable seasonal shopping patterns.
- Evaluated multiple statistical models to forecast future sales, specifically measuring the impact of major market disruptions like the 2020 pandemic and the speed of subsequent recoveries.

Housing Market Analysis R, RStudio

- Investigated the relationship between house and apartment prices across Melbourne and Sydney to determine if different city markets move together in the long run.
- Developed models to perform impact analysis, measuring how local property prices react over time to unexpected market shocks or specific policy changes like urban rezoning.

Finance API Python, FastAPI, BeautifulSoup, PyTest

- Architected a high-speed tool for stock data retrieval, using multithreading to scrape financial statements from multiple sources simultaneously for a 5x increase in speed.
- Engineered a Google Sheets integration used to manage and track \$15,000 in real-world investments, ensuring system stability through automated unit testing.

Technical Skills

Languages: Python, R, Typescript, C, C++17, SQL, HTML/CSS

Frameworks: Flask, FastAPI, PyTest, Vue3, Nuxt3, JUCE

Developer Tools: Git, Docker, RStudio, Linux, Vercel, GCP

Libraries: BeautifulSoup, Pandas, Matplotlib, Numpy/Scipy, Pytorch, Tidyverse, Fable, Cplex, Gurobi, SCIP

Academic Concepts: Analysis, Linear Algebra, Probability, Optimisation and OR, Graph Theory, Forecasting, Statistical Modelling, Econometric Theory